

HOW DO CLOUDS DISSIPATE?

METEOROLOGIST JEFF HABY

theweatherprediction.com

Clouds are composed of liquid suspended water droplets in about a 100% RH environment. **The three primary ways that clouds dissipate is by (1) the temperature increasing, (2) the cloud mixing with drier air, or (3) the air sinking within the cloud.**

(1) When the temperature increases, the air has a higher capacity to evaporate liquid water. When a cloud's temperature increases, evaporation occurs and reduces the liquid moisture content of the cloud. A cloud can be warmed by solar radiation and longwave emission from the earth's surface. Daytime heating increases the capacity of the air to evaporate liquid water. Low clouds such as fog and low stratus are often dissipated due to daytime heating, especially if a cap exists aloft. Daytime heating's power to erode clouds depends on the sun angle (depends on season), the cloud thickness and the overall stability and lift present in the troposphere.

(2) A cloud does not remain perfectly adiabatic (*see tutorial below on adiabatic process*). Some environmental air does mix into the cloud mass. If a cloud is no longer developing and not adding additional condensational moisture, the drier environmental air will gradually erode the cloud. Instead of having a sharp very defined appearance, after mixing with environmental air the cloud will look wispy with edges that are not well defined. This process is called entrainment. During entrainment, drier air incorporates itself into the cloud and induces evaporation.

(3) When air sinks, it warms adiabatically. Again, warming will induce evaporation and erosion of the cloud. This can occur when dynamic sinking mechanisms instigate or increase over a cloud or cloud field. Dynamic sinking mechanisms include low level CAA, NVA, low level divergence, and downslope flow.

HOW DO CLOUDS DISSIPATE? Often, more than one of these processes mentioned above occurs simultaneously to erode a cloud or cloud deck.

ASSUMPTIONS OF ADIABATIC PROCESS

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An adiabatic process is one in which when air rises or sinks there is no exchange of mass, moisture and momentum between the parcel and the environment. What is meant by this? It means the air does not mix. Think of a balloon. The air inside the balloon is trapped and can not mix with the air outside the balloon. The adiabatic process gives us important

thermodynamic information such as: a) unsaturated air cools at the dry adiabatic lapse rate when rising and warms at the same rate when sinking (10 C/km), b) saturated air cools at the wet adiabatic lapse rate when rising, c) the dewpoint lapse rate is a decrease of 2 C/km in rising unsaturated air and increases at same rate when sinking and d) the dewpoint falls with the temperature as saturated air rises.

There are several important points of adiabatic theory I want to go over. The first is time. The longer a parcel of air is exposed to the environment then the more mixing that will occur. Thus an adiabatic process works best in shorter time scales (such as a convective updraft). Once a parcel is exposed to the surrounding environment beyond several hours there will be considerable mixing between the parcel and the environment. This contaminates an adiabatic process. Even in short time scales the adiabatic process will be somewhat contaminated. Unlike a balloon, a parcel of air does not have a membrane separating the parcel air from the environmental air. Think of a helium balloon. When you first buy the balloon it is the most buoyant but over time it eventually will not rise since air has escaped from the balloon and there has been mixing with the air outside the balloon. If drier, more moist, warmer or cooler air is mixed into a parcel it will change the thermodynamic characteristics of that parcel. The parcel has been contaminated so that adiabatic theory will have some error.

The second point I want to go over is the Earth's surface. While air motions are more smooth and less mixed above the planetary boundary layer, air motions near the surface are influenced by friction and this causes a mixing of air (wind gusts, convective eddies, turbulence). These turbulent processes erode the adiabatic process over time. Thus a parcel of air near the surface will mix out with the environment quicker than a parcel of air above the planetary boundary layer. Also, surface heat fluxes and surface evaporation contaminate an adiabatic process. It is easier for moisture to be added or subtracted or heat added or subtracted to a parcel of air if it is exposed to the Earth's surface.

The third point is that an adiabatic process is a theory that only works perfectly under ideal conditions. The atmosphere is not ideal for adiabatic theory. The term adiabatic approximation is sometimes used to make you aware that this process is a theory that does not work perfectly in the real atmosphere. However, it works well enough so that meteorologists are able to use it to gain insight into how thermodynamic characteristics of the air will change as the air rises or sinks. The theory can be used to help forecast thunderstorms, temperature changes in the troposphere and (in)stability of the atmosphere.